ACTA ASTRONAUTICA – FORMAL SUBMISSION COVER LETTER

Date: October 26, 2026
Submission Package Version: 30.0
Manuscript Title: Transmedium Hypersonic Seawater MHD Vehicle Physics: Unified Applied Engineering Specification and Defensive Bounding Analysis
Target Journal: *Acta Astronautica* (Traditional No-Fee Subscription Publishing Model)

To the Editor-in-Chief and Editorial Board of *Acta Astronautica*,

I am pleased to submit the master archival version (Package Version 30.0) of our manuscript titled **"Transmedium Hypersonic Seawater MHD Vehicle Physics: Unified Applied Engineering Specification and Defensive Bounding Analysis"** for publication as a Research Article in *Acta Astronautica*. We explicitly select the traditional no-fee subscription publishing model.

This submission presents a comprehensive, mathematically verified multi-physics engineering architecture for an autonomous transmedium vehicle operating seamlessly across deep-ocean submersion (4,000 m depth, 40.0 MPa hydrostatic pressure) and atmospheric high-supersonic flight (Mach 3.5 to 4.5).

Key Production Updates included in Version 30.0:

1. **Figure 2(a) MHD Thrust Vertex Adjustment:** Calibrated the electrodeless AC MHD propulsion thrust parabola to touch exactly 110.0 kN at optimal slip ratio $s = 0.33$.
2. **Figure 2(b) Electromagnetic Skin Depth Scale Fix:** Fully evaluated physical constants for seawater MHD induction, confirming $\delta = 0.15$ m at 25 kHz.
3. **Figure 4(a) Tsai-Wu Stress Coordinate Sign Inversion:** Corrected the operating stress point marker to Quadrant III (-158.5 MPa, -110.0 MPa), accurately reflecting physical hydrobaric compression.
4. **Complete Mass Allocation & Stability Closure:** Verified 100.00% closure of the 20,000 kg Tier 3 vehicle mass budget, 3D FEA grid convergence ($GCI = 0.42\%$), and Input-to-State Stability via Lyapunov candidate eigenvalues $\lambda(P) \in \{1.43, 2.06, 3.31\}$.

This manuscript is an original work, has not been published elsewhere, and is not currently under consideration by any other journal. All graphics and numerical metrics have undergone final quality assurance and visual audit.

Sincerely,

Jacob James Rasmussen
Independent Researcher — CVTC STEM Division, Chippewa Valley Technical College
ORCID: 0009-0009-7084-6523
Email: rj553483@gmail.com / jrasmussen38@student.cvtc.edu